

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-5684 in Male Participants With Severe Renal Impairment and With End-stage Renal Disease
Short Title:	A Study to Evaluate the Effect of MK-5684 in Male Participants With Severe Renal Impairment (RI) and With End-stage Renal Disease (ESRD) (MK-5684-010)
Protocol Number:	957490228645037065
NCT Number:	NCT06814132
Sponsor Name:	Merck
Investigational Product:	cortisone
Phase:	PHASE1
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Area Under the Concentration Versus Time Curve from 0 to Infinity (AUC0-inf) After a Single Dose of MK-5684: Blood samples will be collected to determine the AUC0-inf of MK-5684 in plasma. 2. Area Under the Concentration Versus Time Curve From 0 to the Time of the Last Quantifiable Sample (AUC0-last) of MK-5684: Blood samples will be collected to determine the AUC0-last of MK-5684 in plasma. 3. Area Under the Concentration Versus Time Curve From 0 to 24 Hours After Dosing (AUC0-24) of MK-5684: Blood samples will be collected to determine the AUC0-24 of MK-5684 in plasma.
Secondary Obj.:	1. Number of Participants who Experience an Adverse Event (AE) 2. Number of Participants who Discontinue Study Intervention Due to an AE 3. Dialysis Clearance (CLD) of MK5684 Based on Plasma

2.2 Background & Rationale

The goal of the study is to learn what happens to levels of MK-5684 in people with severe renal impairment and end-stage renal disease versus a healthy person's body over time. Researchers will compare what happens to MK-5684 after hemodialysis in people with severe renal impairment and end-stage renal disease versus healthy people.

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE1, Status: COMPLETED
Target N:	24

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

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Inclusion criteria include, but are not limited to:

All Participants

- * Is a continuous non-smoker or moderate smoker (? 10 cigarettes per day or equivalent) for at least 3 months prior to the first dosing
- * Has a body mass index (BMI) ? 18.0 and ? 42.0 kg/m² at the screening visit

Participants with severe renal impairment (RI) (Group 1):

- * Has severely impaired renal function as determined by estimated glomerular filtration rate (eGFR) using the 2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation

Participants with end-stage renal disease (ESRD) (Group 2):

- * Has ESRD maintained on stable outpatient regimen of intermittent high-flux hemodialysis (HD) (at least 3 times per week) for a minimum of 3 hours per dialysis session, using a complication free well-maintained arteriovenous fistula or arteriovenous graft, for at least 3 months prior to first dosing and is expected and planning to continue HD during the study and at least up to the follow-up visit

Healthy Control Participants (Group 3):

- * Has normal renal function

4.2 Exclusion Criteria

Exclusion criteria include, but are not limited to:

All Participants:

- * Has a history or presence of any of the following: Adrenal insufficiency; Hepatic impairment; Clinically significant hypotension; cardiac arrhythmia; cardiac conduction abnormalities or recurrent unexplained syncopal events; Second- or third-degree atrioventricular (AV) heart block (except in participants with a functional artificial pacemaker); Clinically significant sick sinus syndrome; Any systemic fungal infection; Hypothyroidism; Chronic infection
- * Has a first-degree relative with multiple unexplained syncopal events, unexplained cardiac arrest, or sudden cardiac death, or has a known family history of an inherited arrhythmia syndrome (including Brugada syndrome)
- * Has a history of cancer (malignancy)
- * Is positive for human immunodeficiency virus (HIV), hepatitis B surface antigen (HBsAg), or hepatitis C virus (HCV)

Participants with Severe RI (Group 1):

- * Has a history or presence of renal artery stenosis

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- * Has had a renal transplant
- * Has ESRD requiring HD

Participants with ESRD (Group 2):

- * Has required frequent emergent HD (? 3) within a year prior to the first dosing
- * Has a history or presence of renal artery stenosis.
- * Has had a functioning renal transplant within the past 2 years

11. Study Identifiers & Governance

Primary ID: 957490228645037065

Registry Number: NCT06814132

Sponsor: Merck

Study Title: A Study to Evaluate the Effect of MK-5684 in Male Participants With Severe Renal Impairment (RI) and With End-stage Renal Disease (ESRD) (MK-5684-010)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____